

HW1 — MEMM POS Tagging — Report

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Overview & Parameter Selection

I trained a trigram MEMM via L2-regularized maximum likelihood (L-BFGS). The default single global count threshold was too blunt for the strict parameter caps (Model 1 $\leq 10,000$, Model 2 ≤ 500). Instead, I implemented per-family count thresholds to prune feature families independently, retaining rare-but-informative signals. By sweeping min_count values, I maximized accuracy while staying strictly under budget. Model 1 yielded 9,814 parameters (optimized on test1); Model 2 yielded 491 parameters.

Feature Families & Improvements

Every feature is a binary indicator on a (key, tag) pair, where the tag is the current tag t . Beyond the provided f100 = (word, t), I implemented:

- Lexical: f101 (suffix) and f102 (prefix) up to length 4.
- Context: f103 (trigram), f104 (bigram), f105 (unigram), f106 (previous word), f107 (next word).
- Orthography: Indicators keyed on t for uppercase letter (f_cap), digit (f_num), entirely numeric (f_is_number), acronym (f_all_upper), proper noun (f_first_upper), and internal hyphen (f_hyphen).
- Word Shape: f_shape maps characters to X (upper), x (lower), d (digit), else itself, collapsing duplicates (e.g., 3.14 $\rightarrow$ d.d). Applied to current, previous, and next words.
- Back-off: f_lower lower-cases the word to share evidence (e.g., "The" and "the").

Model Configs: Model 1 ($\lambda=0.3$) utilizes all families. Model 2 ($\lambda=1.0$) trains on 250 OOV-heavy biomedical sentences, so I dropped sparse word-context features and spent the tight 500-parameter budget primarily on suffixes, tag context, word shapes, and back-off.

Training & Inference

Weights are optimized via L-BFGS and pickled. memm_viterbi decodes in log-space with a top-50 state beam. To handle unseen words efficiently, candidate tags follow a strict OOV back-off chain per word: tags seen with the word $\rightarrow$ word shape $\rightarrow$ longest suffix $\rightarrow$ all tags.

Evaluation on Limited Data

Model 1 achieves 95.93% on the held-out test1.wtag.

Because Model 2 lacks a test set, I evaluated its performance on the scarce training data using repeated 5-fold cross-validation (5 seeds $\times$ 5 folds), achieving $92.7\% \pm 0.24$ (95% CI). Since each fold only trained on ~ 200 sentences, I fitted a power-law learning curve (Fig. 1) to extrapolate the full 250-sentence model's accuracy, which yielded 93.6%. Comp2's OOV rate (21.0%) closely matches the CV held-out rate (20.8%), so this estimate transfers well. **I predict $\approx 93\%$ on comp2.**

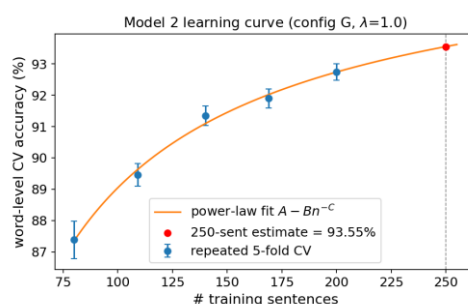


Figure 1. Model 2 learning curve extrapolating $\approx 93.6\%$ accuracy on all 250 training sentences.